

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

1. Multi-Factor Authentication (MFA)

Implement MFA for all employee and administrator accounts. MFA requires users to provide an additional authentication factor beyond a password before gaining access to systems and data.

2. Strong Password Policy

Implement a strong password policy requiring complex passwords, regular password updates, and the prohibition of password sharing. Default passwords should be replaced immediately with unique and secure credentials.

3. Firewall Rules and Port Filtering

Configure firewall rules to filter incoming and outgoing network traffic. Only approved services and ports should be allowed, while unnecessary traffic should be blocked.

Part 2: Explain your recommendations

1. Multi-Factor Authentication (MFA)

MFA is effective because it provides an additional layer of security beyond passwords. Even if an attacker obtains a user's password through guessing, credential theft, or brute force attacks, they would still need the second authentication factor to gain access. MFA should be enforced continuously for all privileged and user accounts.

2. Strong Password Policy

Strong password policies reduce the likelihood of successful brute force attacks and unauthorized access. Requiring complex passwords and eliminating default passwords helps protect critical systems such as databases and administrative accounts. Password policies should be reviewed regularly and enforced whenever accounts are created or modified.

3. Firewall Rules and Port Filtering

Firewall rules help protect the network by controlling which traffic is permitted

to enter and leave the environment. Port filtering reduces the attack surface by blocking unnecessary services and restricting access to authorized traffic only. Firewall configurations should be monitored and reviewed periodically to ensure they remain effective against emerging threats.